Team Batties: Trevor Boshnack, Eva Gottesfeld, Madeline Mezynski, Eli Pate, Diya Sheth
Client: Cornell CALS Extension, E&J Gallo Winery, and National Grape

Context and Problem Statement

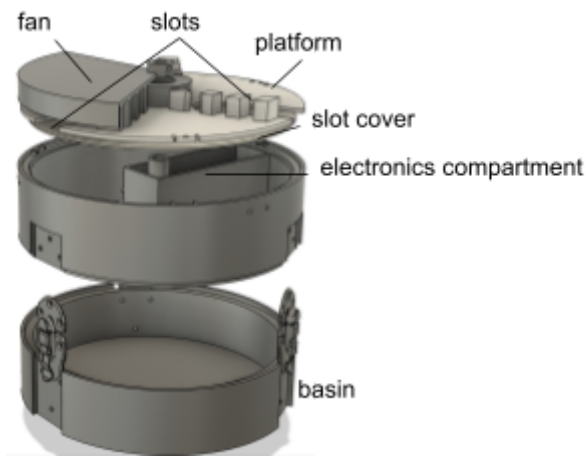
Spotted lanternflies (SLF) pose a significant threat to the grape industry, for as many as 400 SLF can feed on a single grape vine (Cornell CALS, n.d.). Once they enter a vineyard, they are difficult to eliminate due to the dense foliage, their unpredictable movement, and their fast reaction time relative to humans. Over time, repeated feeding can severely damage vines and result in harvest losses of up to 90% (Pfeiffer et al., 2022).

If SLF continue to feed on the grapes before harvest, the overall crop yield may be significantly reduced. Preventing SLF from reaching the vineyards directly reduces both crop damage and the risk of grape juice contamination during harvest. For this reason, the goal is to capture SLF as they migrate from the tree of heaven toward the vineyards. The device is designed to be portable, low-maintenance, and independent of a wired power source so it can be deployed wherever it is needed.

Final Prototype and Application

The prototype is a motorized spinning trap inspired by an electronic flytrap. A bristled fan rotates around a central shaft and pushes the SLF through slots in the platform into a liquid-filled basin below. As the fan passes over the insects, bristles contain them from multiple sides, leaving the basin as the only exit path. To prevent SLF from escaping through other openings, a rotating plate beneath the platform keeps all slots covered except the one directly under the fan. Because the system is powered by a motor, the trap can operate without direct monitoring.

When the client wishes to empty out the trap, they can use the latches to open and clean out the basin. The trap can either be hung from a tree or rail using the hook on the roof, or it can be placed in an open field using the tripod setup on the bottom.



Assembly

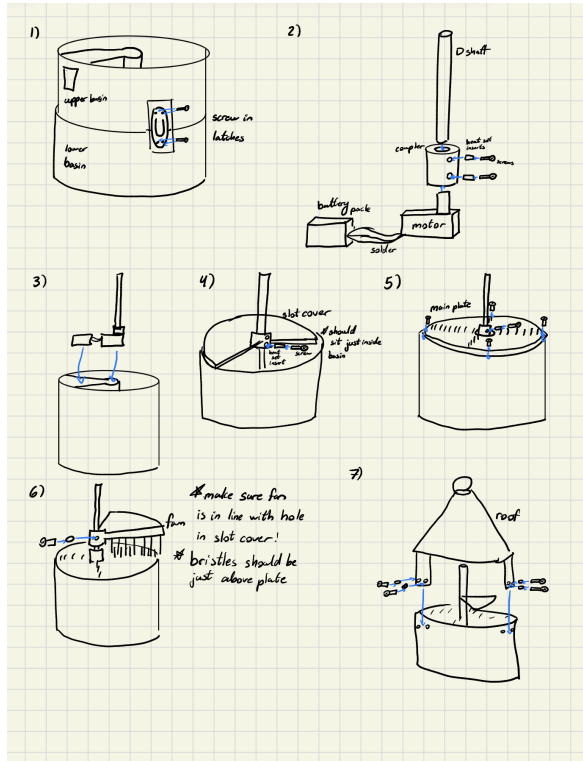
Step 0. Prepare all materials.

a. 3D print the basin, upper and lower basins, fan, slot cover, and roof.
b. Install heat-set inserts into each hole in the 3D printed components.

Step 1. Connect the upper and lower basins by screwing the latches into the designated holes (latch space is indented for clarity).
Step 2-3. Assemble the electronics.

a. Couple the shaft to the motor using a coupler with set screws and set it in the center of the basin.
b. Press fit the switch into the rectangular hole.
c. Attach the red battery wire to a motor terminal, the black battery wire to a switch terminal, and a wire between the switch and the motor. Power on and check that the motor rotates clockwise; if it doesn't, switch motor terminals used. Solder terminals once confident in configuration.

Step 4. Slide the slot cover on the rod to sit just above the lip of the basin. Tighten both set screws in the heat set inserts.
Step 5. Attach the main plate using designated cutouts on the side of the basin, then screw to tighten.
Step 6. Attach the fan to the rod so the bristles are <1-2mm above the main plate by tightening set screws. Ensure the fan covers the area the slot cover does not!
Step 7. Slide the roof onto the rod and screw its arms in.



Conclusion and Recommendation

The goal was to create a trap that can capture SLF without constant human monitoring. Testing showed that the bristles successfully guide objects into the slots, the motor is strong enough to push and crush SLF, and the mechanism shows minimal wear after repeated use. The prototype also remained under budget, making it a promising candidate for future field testing.

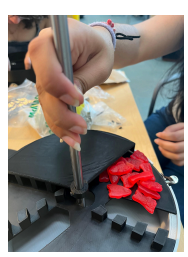
Before field testing, several improvements should be made. We determined the current bristle spacing to be wide enough for smaller SLF to slip through, so reducing the spacing is recommended. A lack of rigidity in the motor mount caused the shaft to sit slightly crooked, so the area around the motor should be reinforced. Finally, future versions should be designed to run on solar power to reduce the need for battery replacement. The next phase of testing should focus on SLF behavior. Future researchers should:

1. Identify the most effective attractant, ideally one that outperforms grapes.
2. Measure the number of SLF captured per day to determine the appropriate basin size and maintenance schedule.
3. Evaluate how quickly SLF respond to the fan mechanism in order to determine the optimal operating speed of the device.

Testing and Results

Success criterion 1: The fan shall carry objects as small as 5mm (SLF-sized objects) into the slots.

The purpose of this test was to determine whether the mechanism could successfully push SLF into the slots. SLF were modeled using halves of Swedish Fish,



Initial #	Avg % remaining
5	0
10	3.3
15	4.5
20	3.4

Table 1. *Remaining Percentage of SLF on the platform after one sweep of the fan.* The right number is the initial amount on the platform.

approximately 5 mm by 5 mm. Varying densities of fish were placed on the platform. After one sweep of the fan, the number of fish remaining was recorded. Average percentages are shown in the figure above.

During testing, any Swedish fish remaining were either pushed off the side of the plate or into the center. Thus, future models should add walls around the platform to keep SLF contained.Some Swedish fish also managed to bypass the bristles at the exhibition, so the bristle size must be decreased.

Success criterion 2: The fan shall move through static bristles on the SLF platform with maximum of 2.356 N*m of torque.

The purpose of this test was to confirm that the motor could operate reliably with minimal maintenance. To simulate a maximum load condition, a Swedish Fish, which is denser than an SLF, was placed between the tines to represent an obstruction. The fan was then pulled through the tines using a force sensor, and the required torque was calculated. In all trials, the required torque was significantly lower than the motor's maximum torque, indicating that the motor is strong enough for the expected operating conditions.

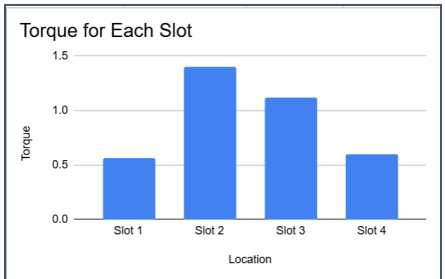


Figure 2. *Torque required to shear an SLF in each slot.* Torque decreased when SLF's sat in the closer slots, except for the farthest slot in which it's easier to shear the SLF

Success criterion 3: The motion of the fan reliably continues for at least 15 consecutive revolutions with the same time per revolution.

Because the machine will likely remain outdoors for several days at a time, it must be able to operate reliably over time. The time for each revolution was recorded to evaluate consistency. Although there were slight variations in speed, no significant trend was observed, suggesting that the fan is well supported throughout and there is minimal binding. Additionally, there were no visible signs of wear or damage after testing.

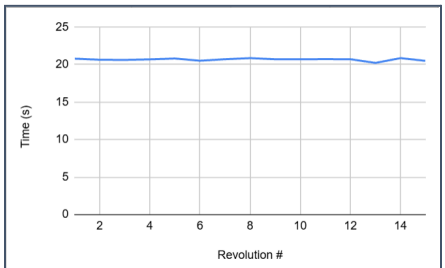
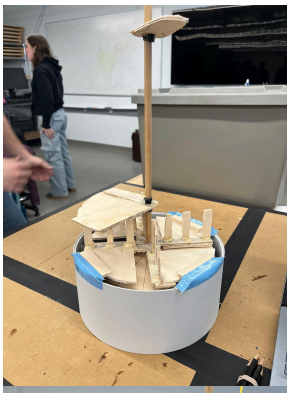


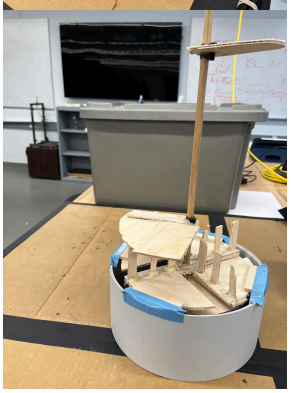
Figure 3. *Period of fan vs number of revolutions.* The period fluctuated slightly, but it remained relatively constant throughout.

Prototype 1

Our first prototype was primarily constructed from wood and was used to model the basic rotating mechanism and the underlying slot-cover system.

We first tested whether the fan could move smoothly through the bristles. During trials without model SLF, misalignment in the wooden bristles caused the force meter to max out several times. This showed that the bristles needed to be more precise, which led us to develop an almost entirely 3D-printed model. Some of the misalignment was also caused by the fan sagging around the shaft, so we made the fan hole taller and a tighter fit to the shaft.





We also evaluated whether friction of the slot cover would become a major issue, especially for the rotating plate beneath the platform. We did not feel significant friction during operation, yet still thought it best to space the slot cover so that it would not contact the components above or below it.

Lastly, we tested whether the bristles could successfully move model SLF, which were represented by foam shreds in this experiment. The test was conducted in a similar way to the experiment used for Success Criterion 1. The prototype was able to move the foam pieces, which suggested that the design concept was feasible. However, nearly 40% of the model SLF remained after testing. In the final prototype, this issue was addressed by adding walls around the platform, lowering the fan relative to the platform, and improving the bristle spacing.

References

Pfeiffer, D.G. et al. (2022) *Spotted lanternfly in virginia vineyards: Lycorma Delicatula (white) (Hemiptera: Fulgoridae).* doi:10.21061/ento-323np.

Spotted lanternfly damage. Cornell CALS. (n.d.). https://cals.cornell.edu/integrated-pest-management/outreach-education/whats-bugging-you/spotted-lanternfly/spotted-lanternfly-damage

Component List

Part (prototype 1)	Specs	Fabrication Details	Cost
Sweeping Rotor	⅛ inch thick balsa wood, quarter of a 10cm radius circle.	Cut out of wood using a bandsaw/hand saw and sanded to the planned dimensions. Teeth cut out of wood using a bandsaw/hand saw and hot glued/gorilla glued onto the disk. Hole drilled in the middle for the dowel.	Free (made from scrap material)
Spinning Open Disk	⅛ inch thick balsa wood, 21cm in diameter.	Cut out of wood using a bandsaw/hand saw and sanded to the planned dimensions. Hole drilled in the middle for the dowel.	$1.75
SLF Platform	⅛ inch thick balsa wood, 22cm in diameter.	Cut out of wood using a bandsaw/hand saw and sanded to the planned dimensions. Teeth cut out of wood using a bandsaw/hand saw and hot glued/gorilla glued onto the disk	$1.75
Rod	⅜ inch dowel	Cut to size	free (from scrap material)
Crank	⅛ inch thick balsa wood, in an oval shape	Cut out of wood using bandsaw and sanded down using sander. Hole drilled in the middle for the dowel. Connected the crank to the dowel with hot glue and zip ties.	$1.75

Electronics Casing	23x23x10cm, PLA	3D Printed by RPL	22.68

Part (prototype 2)	**Specs**	**Fabrication Details**	**Cost ($)**
8632T381	10mm D-shaft	Cut with bandsaw in machine shop	18.55
Shaft coupler	6mm to 10 mm coupling	Amazon	6.99
DC Motor with Gearbox	6mm shaft, 2.356 N*m torque	Amazon, soldered to batteries	15.87
7194K101	Switch	McMaster, soldered to batteries	15.56
Battery Pack	8 batteries, totalling 12V	Amazon	7.99
94459A772	3mm heat set insert	Melted into 3D prints with solder gun	15.46
92196A148	Latch screws	McMaster	7.62
Lower Basin	20cm diameter	3D print–PLA	
Upper/electronics Basin	20cm diameter	3D print–PLA	
Slot Covers	20cm outer, 10mm inner	3D print–PLA	
Fan	10cm radius, 10mm inner diameter, 15mm bristles	3D print–PLA	
Base plate	20cm diameter, 15mm bristles	3D print–PLA	
Roof	20cm diameter	3D print–PLA	

Bill of Materials

Part number	Purpose	List Price	Quantity	Total Price	
8632T381	Main shaft	18.55	1	18.55	
Amazon	6mm to 10mm Shaft Coupler	$7.00	1	$7.00	
Amazon	Motor	$15.87	1	$15.87	
7194K101	Switch	$15.56	1	$15.56	
Amazon	Battery pack	$8.00	1	$8.00	
94459A772	connect fan pieces to shaft	$7.73	2	$15.46	
1766A2	Connect basin and electronics basin	$15.20	1	$15.20	
92196A148	Screws for latches	$7.62	1	$7.62	
90480A007	Nuts for latch screws	$1.57	1	$1.57	
94459A769	Heat Set Inserts M3	$8.64	1	$8.64	
PLA Filament Spool	3D printed parts	$12.00	1	$12.00	
Prototype #1	Proof of concept parts	$5.25	1	$5.25	
Total Budget:	350		**Total Cost:**		130.72